Can we hear depression and anxiety in how people speak?

A plain-language summary of the MINNs speech experiments · October 2026

We took the voice recordings of the participants, cut each recording into many short windows, and used large pre-trained speech models to turn every window into a list of numbers. Small classifiers then learned which windows go with a positive screen. We compared 4 speech models, window lengths, ways of combining windows, and the multiple-instance networks (MINN) from Alsarrani et al. 2025.

Ranking (AUC)

best ≈ 0.67 on known sites
≈ 0.66 on unseen sites

Better than

the earlier pipeline (0.57–0.61)
and age/gender/education (0.60)

But

F1 only 0.18–0.30 (flag-everyone 0.21);
MINNs add little over the probe

Papers followed: Alsarrani et al., ICMI 2025 (MINN, "punctual vs continuous"); de Gennes et al. 2024 (probing speech foundation models).

1. The data and how we test

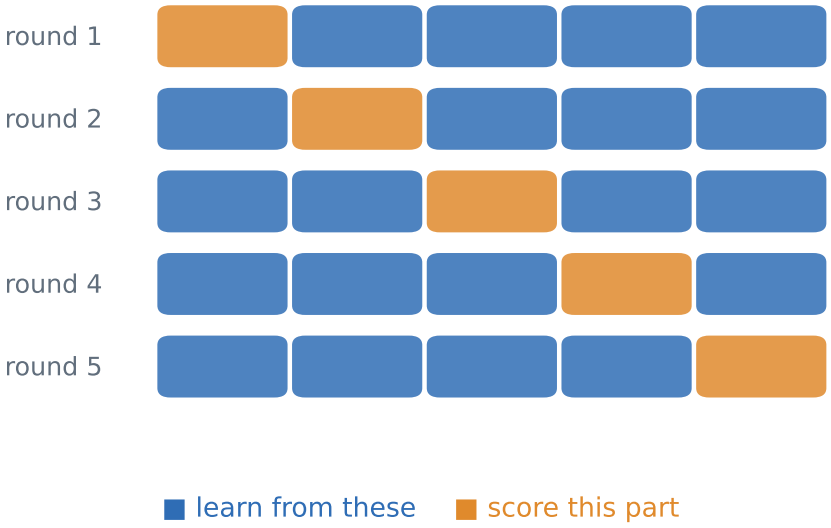
Same participants, labels and splits as the mh-ML-pipeline

674 participants with a voice recording and a screening label



63 of the 550 (11%) screen positive. The 124 test participants (14 positive) have NOT been looked at – they are saved for one final check.

5-fold cross-validation



Every participant is scored once, by a model that never saw them.

Two ways of making the 5 folds

Stratified split (known sites): every fold has the same mix of label, cohort, illness, site, language, education and age. Each site appears on both sides → new people at familiar sites.

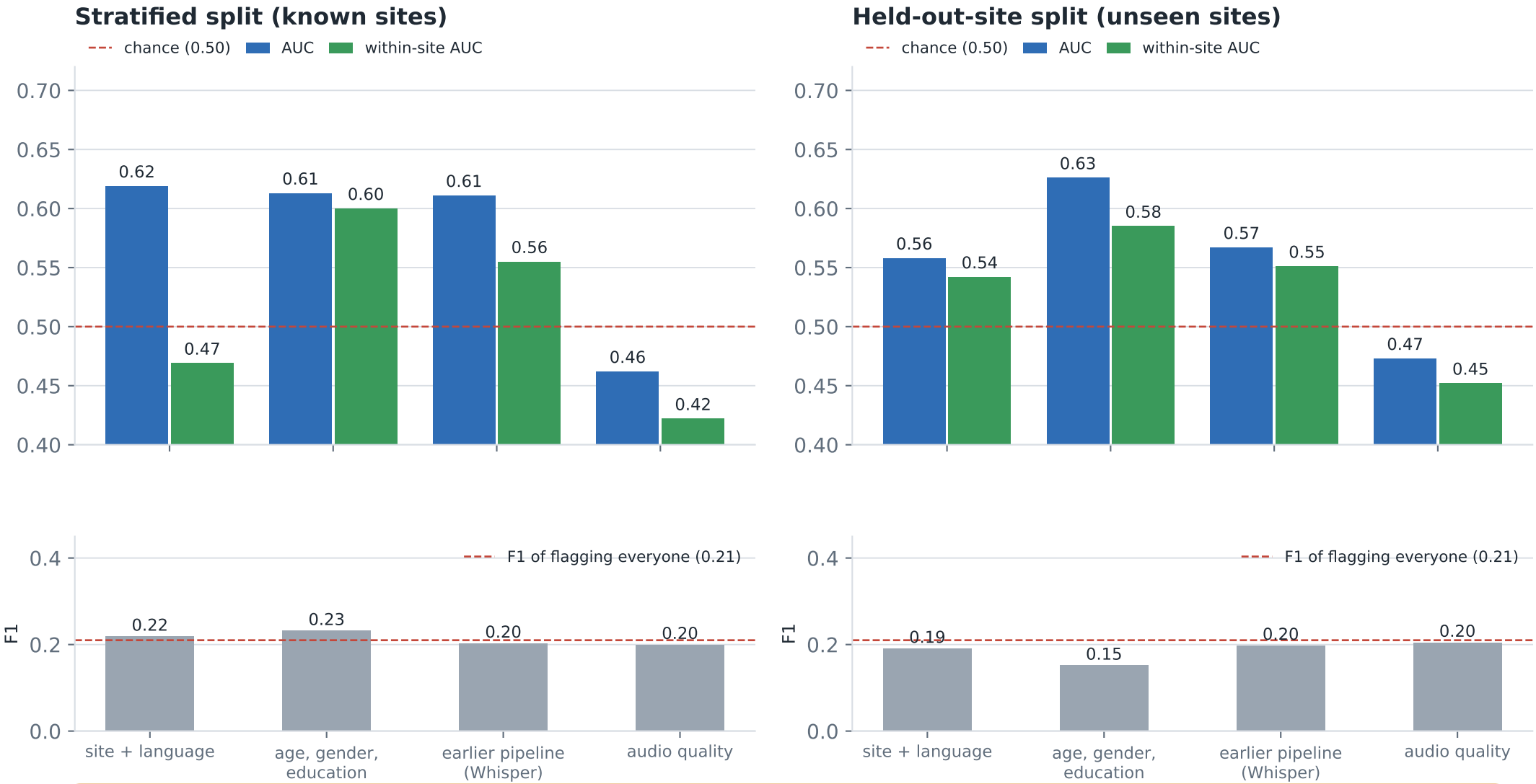
Held-out-site split (unseen sites): whole sites are kept out of learning, folds balanced on the label only → a site never seen before.

Three scores on every chart

AUC – ranking quality, no threshold: 0.5 = coin flip, 1.0 = perfect.
Within-site AUC – the same, but people only compared with others from their own site, so site differences can't help (see page 3).
F1 – after picking a yes/no cut-off (tuned for F2 on the other folds). Calling everyone positive already gives F1 ≈ 0.21.

2. First, a trap to avoid: "cheating" signals

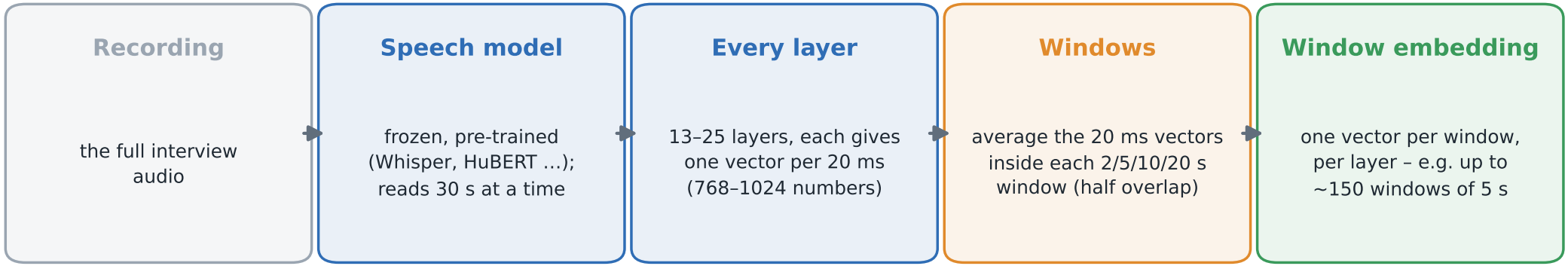
Stage 0 – how well can we do WITHOUT the new speech models?



Some sites simply have more positive participants: site + language alone gets AUC 0.62 without hearing a word. Within-site AUC removes this (0.47). Age, gender and education are a real bar: within-site 0.60 / 0.59 (women screen positive about twice as often, 15% vs 7%). A speech model has to beat ≈ 0.60, not 0.50.

3. From a recording to window embeddings

The same for every speech model



"Whole recording" (what the earlier pipeline did)

Instead of windows, average ALL 20 ms vectors of the recording into one vector per person, then train a classifier on that one vector.

Simple, but any short moment that matters is diluted by minutes of ordinary speech.

How "best" is chosen

For each speech model we tried every combination of layer × window length × pooling and kept the one with the highest AUC – layer, window and pooling are chosen together, separately for each split.

Labels like "L23 · 5 s · max" under bars name that choice. Best-of-many is slightly optimistic.

4. Two ways to learn from windows: probe vs MINN

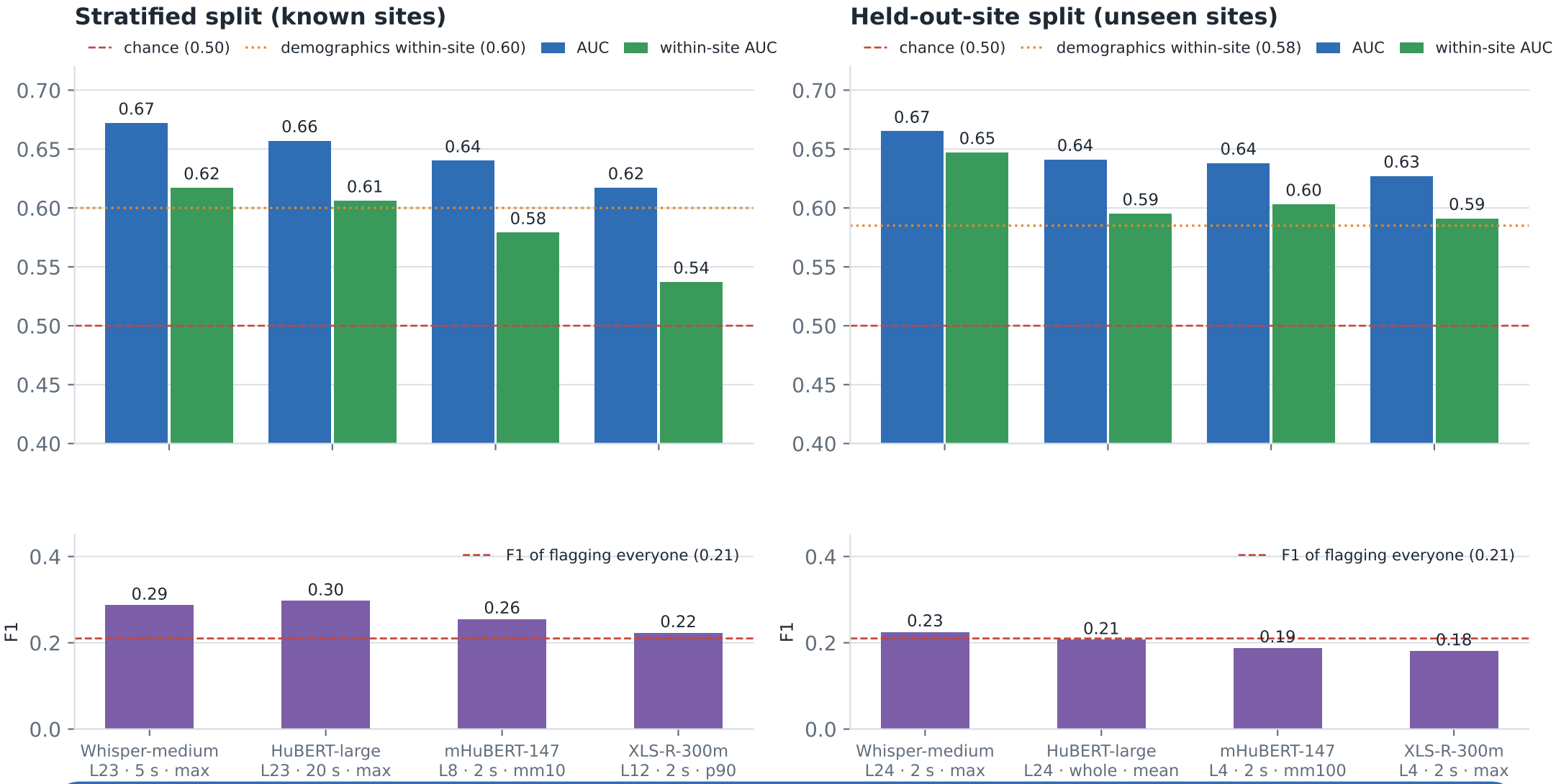
Both use the same frozen window embeddings – they differ in how they are trained

	Training	Predicting
Probe	Every window gets its person's label (a positive person's windows all count as positive). A logistic regression learns to score single windows. Pooling plays no part in training.	Window scores are combined into one score per person. We try several poolings on the same trained model: max, mean, min, p90, vote, mellowmax (mm).
MINN (multiple-instance network)	Window scorer → pooling → one score per person, trained together on the person's label only. The model can learn that only some windows matter. Scorer: linear, or a small neural network (32–128 units).	The pooling it was trained with (max, mean or lse – a soft maximum). Variants: mi-Net (score windows, then pool), MI-Net / attention (pool features, then score).

"Probe + max" and "linear MINN + max" have exactly the same form – one weighted sum per window, then the maximum.
Only the training differs, so comparing them tests one question: does learning through the pooling help? (page 8)
Why pooling matters: a sign of low mood may appear only in a few moments ("punctual"); an average hides them, the maximum keeps them.

5. Which speech model hears it best?

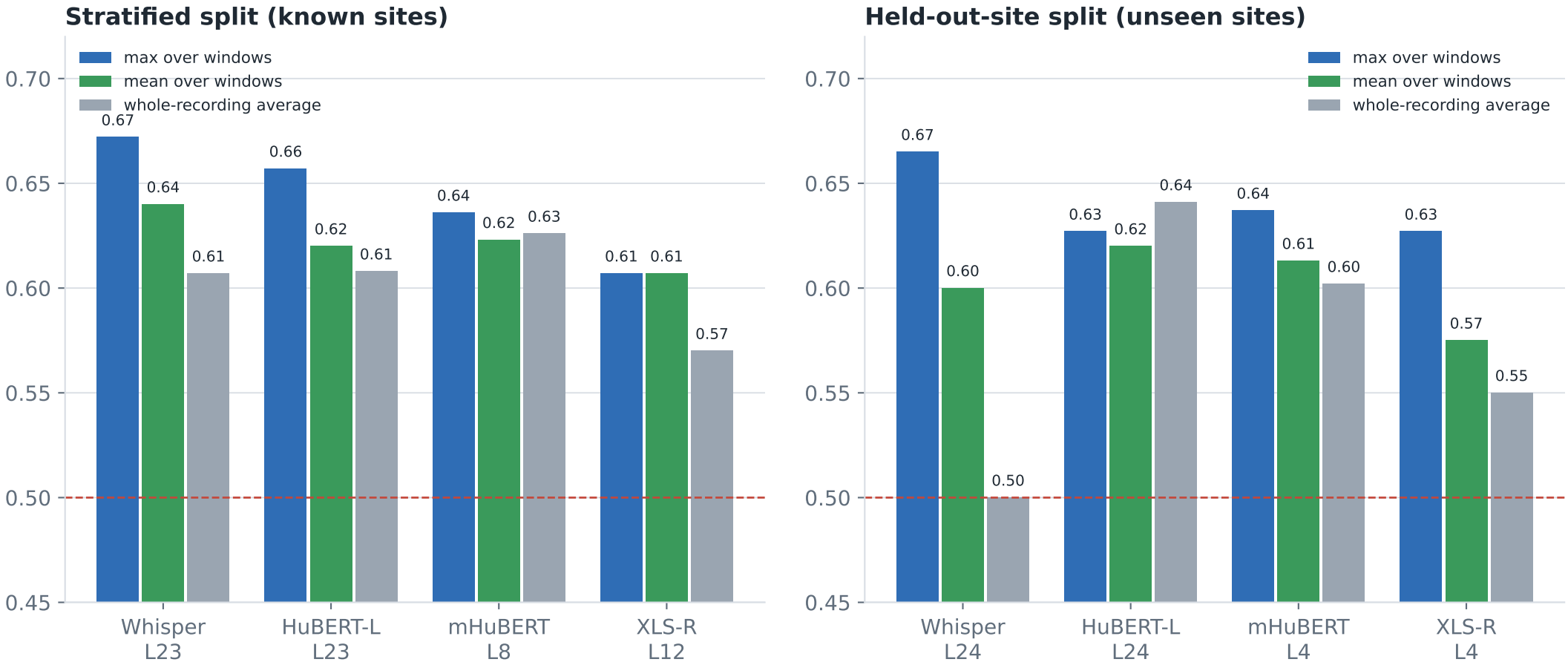
Stage B – probe on each speech model, best layer · window · pooling per split



Whisper-medium is best on both splits, with one of its last layers (23–24) and max pooling, and is the only model above the demographics bar within site on both splits. XLS-R's middle layers (15–18) fall below chance on unseen sites. F1 is low everywhere (0.18–0.30); on unseen sites it is barely above flagging everyone (0.21).

6. Does keeping the strongest windows help?

AUC of the probe at each model's best layer: max vs mean over windows vs one whole-recording average

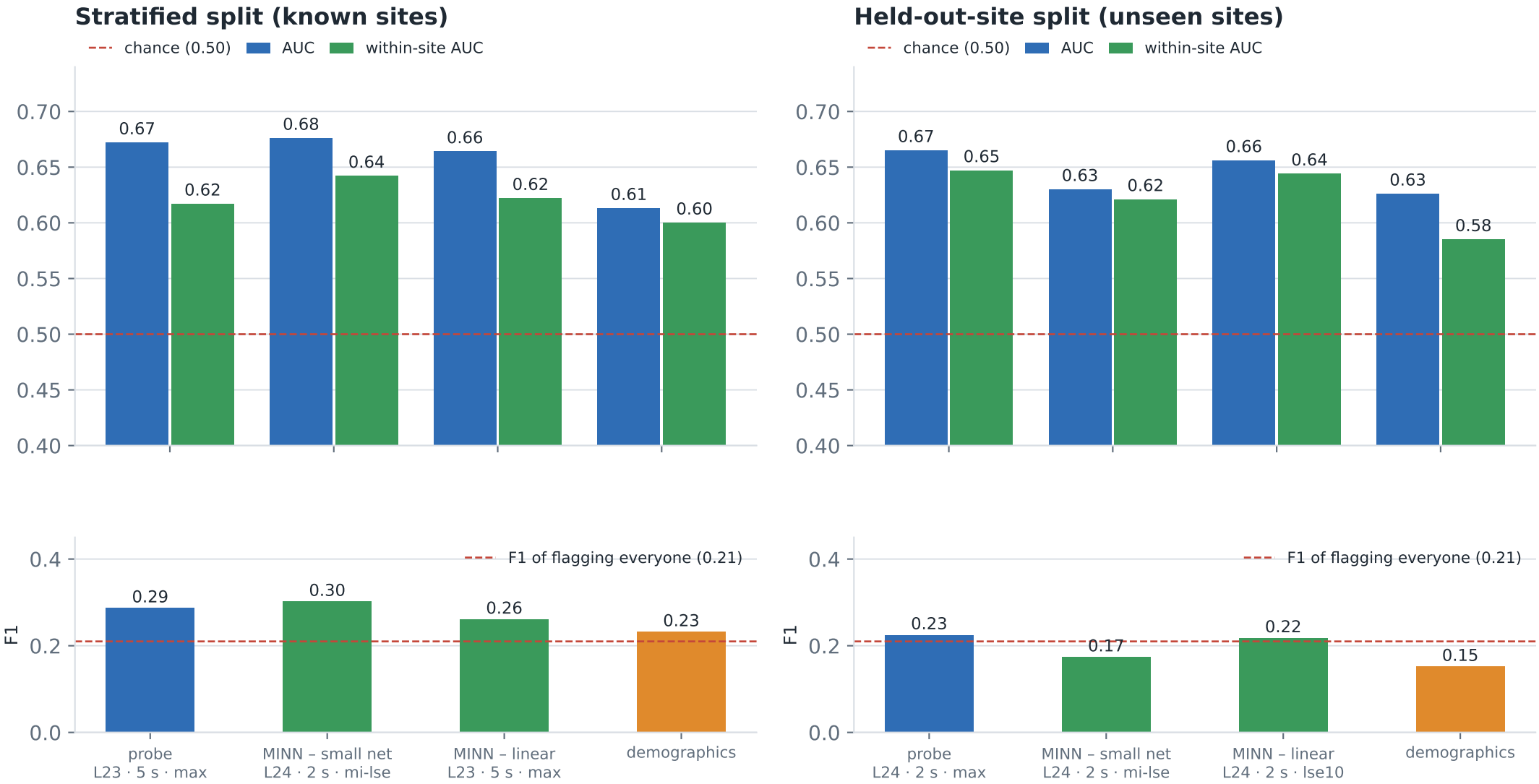


- Whisper: max clearly beats mean (+0.03 known, +0.065 unseen sites); its whole-recording average is chance (0.50) on unseen sites.
- HuBERT-L: max helps on known sites (+0.04), but on unseen sites its whole-recording average is its best result (0.64).
- mHuBERT: small gain from max (+0.01 / +0.02). • XLS-R: no gain on known sites, +0.05 on unseen sites.

→ "Punctual" signal (max wins) is clear for Whisper, the best model; it is not universal.

7. Do the multiple-instance networks (MINN) help?

Stage C – Whisper-medium, best setting of each approach per split



All three land at AUC ≈ 0.63–0.68; the gaps (≤ 0.035) are smaller than the fold-to-fold spread (±0.03 to ±0.12), so no MINN is reliably better than the probe. The small network wins slightly on known sites but loses on unseen sites. MI-Net and attention variants fell to ≈ chance on unseen sites; HuBERT-L MINNs were worse than its probe.

8. Everything in one table

Each row: best setting per split (5-fold CV on the 550). Darker green = better; red = at or below chance / flag-everyone.

	Stratified (known sites)			Held-out site (unseen sites)		
	AUC	within-site AUC	F1	AUC	within-site AUC	F1
Site + language only	0.62	0.47	0.22	0.56	0.54	0.19
Age, gender, education	0.61	0.60	0.23	0.63	0.58	0.15
Audio quality	0.46	0.42	0.20	0.47	0.45	0.20
Earlier pipeline (Whisper, whole recording)	0.61	0.56	0.20	0.57	0.55	0.20
Whisper – probe	0.67	0.62	0.29	0.67	0.65	0.23
Whisper – MINN small network	0.68	0.64	0.30	0.63	0.62	0.17
Whisper – MINN linear	0.66	0.62	0.26	0.66	0.64	0.22
HuBERT-large – probe	0.66	0.61	0.30	0.64	0.59	0.21
mHuBERT-147 – probe	0.64	0.58	0.26	0.64	0.60	0.19
XLS-R-300m – probe	0.62	0.54	0.22	0.63	0.59	0.18

Bold rows = this project. F1 shaded red when ≤ 0.21 (no better than calling everyone positive). Each row's setting is the best of many, so values are slightly optimistic.

9. Honest caveats and next steps

Caveats

- Small numbers: about 13 positive people per fold, so AUC moves ±0.03–0.12 between folds.
- We tried many settings and report the best; the true performance is probably a bit lower.
- F1 is low (0.18–0.30). As a yes/no screener the model is not usable yet – AUC shows a real but weak signal.
- Speech may partly pick up gender, which is linked to the label – not yet ruled out.
- Nothing has been checked on the 124 test people.

Suggested next steps

1. Speech vs demographics: does Whisper add anything on top of age, gender and education?
2. Remove site effects from the speech features (normalise within each site) before scoring.
3. Pick ONE final model with a rule fixed in advance, retrain on all 550 people and score the 124 test participants once.

Glossary

Window (instance) – one short slice of a recording. Bag – all windows of one person.
Layer – one processing stage inside the speech model (L23 = layer 23). Embedding – the list of numbers describing a window.
Probe – a simple classifier on frozen embeddings, trained window by window. MINN / MIL – learns from whole bags labelled only as a whole.
Pooling – how window scores become one person score: max, mean, min, p90 (90th percentile), vote, mm/lse (soft maximum).
mi-Net – score each window, then pool. MI-Net / attention – pool the window features, then score.

Code: src/minns/ (extract, probe, mil, baseline, evaluation) · Cluster jobs: scripts/cluster_{extract,probe,mil}.sh · Results: ~/MIL/outputs/results/ on the cluster